

Revisiting The Normal Distribution

- A _____ random variable is a variable that can take on any value within a specified domain.
 - Every interval within the domain has a _____ associated with it.
- A continuous random variable with a normal distribution is commonly used to describe populations.
- The area under a normal curve, over a given interval represents the probability that a particular x-value lies in that interval:

$P(X < x_a)$	$P(x_a < X < x_b)$	$P(X > x_b)$
The probability a randomly selected variable is less than x_a	The probability a randomly selected variable is between x_a and x_b	The probability a randomly selected variable is greater than x_b

Remember:

- $P(X = x_a)$ _____ for any continuous random variable
- $P(X \geq x_b)$ is the same as _____ since individual values in a continuous distribution are insignificant

Solving Normal Random Variables

	Example: The heights of young women follow a Normal Distribution, with a mean height of 64 inches and a standard deviation of 2.7 inches. If you were to choose one woman at random, what is the probability she is less than 63 inches?	
Step 1	Communicate what your random variable is and the distribution it follows.	
Step 2	Write down your probability statement and convert into a z-score.	

Step 3	Draw your curve and find your probability using technology. TI 84: 2nd → VARS normalcdf(lower, upper, μ, σ)	
Step 4	Write a sentence communicating the answer.	

Ex 1) The heights of young women follow a Normal Distribution, with a mean height of 64 inches and a standard deviation of 2.7 inches. If you were to choose one woman at random, what is the probability she is greater than 66 inches tall?

Ex 2) If you were to choose one woman at random, what is the probability she is between 64.5 inches and 70 inches tall?

Combining Normal Random Variables

- Shape: Combining two or more normal distributions will result in a combined _____
- Center: $E(X + Y) = \underline{\hspace{2cm}}$ and $E(X - Y) = \underline{\hspace{2cm}}$
 - $E(X)$ is also represented as μ_X
- Spread: $\sigma_{X+Y} = \underline{\hspace{2cm}}$ and $\sigma_{X-Y} = \underline{\hspace{2cm}}$ – always add variances!!
 - The random variables must be independent!
- Why are we combining?
 - We combine multiple normal random variables to _____

Adding Normal Random Variables Example

In a 4x100 track relay, four team members each run 100 yards, and their combined total time is their finishing time. Suppose the times taken by each runner to complete their respective legs of the relay are independent of each other and follow a normal distribution with the following parameters:

Runner 1: Mean = 12 seconds, Standard Deviation = 1 second

Runner 2: Mean = 11 seconds, Standard Deviation = 0.8 seconds

Runner 3: Mean = 10.5 seconds, Standard Deviation = 1.2 seconds

Runner 4: Mean = 11.5 seconds, Standard Deviation = 1 second

What is the probability that the total time taken by the team to complete the relay race is between 42 and 46 seconds?

Subtracting Normal Random Variables Example

You are managing a manufacturing process that involves two sequential steps (steps taken at the same time). The two processes work independent of each other and the time taken to run each step can be modeled with a normal random variable representing the processing time. Process 1 has a mean processing time of 25 minutes and a standard deviation of 5 minutes, while Process 2 has a mean processing time of 20 minutes with a standard deviation of 4 minutes. You need the difference in processing time to be less than 10 minutes for a critical operation. What is the probability that you meet this requirement?